

Macro Copilot

Tool Catalogue and Product Overview

Deterministic analytics and transparent workflow orchestration for discretionary macro research

Problem

Macro pods have abundant data, strong market intuition, and skilled analysts. The recurring bottleneck is research latency: finding the right sources, running standard analytics, reconciling conventions, and turning results into an investment-oriented read.

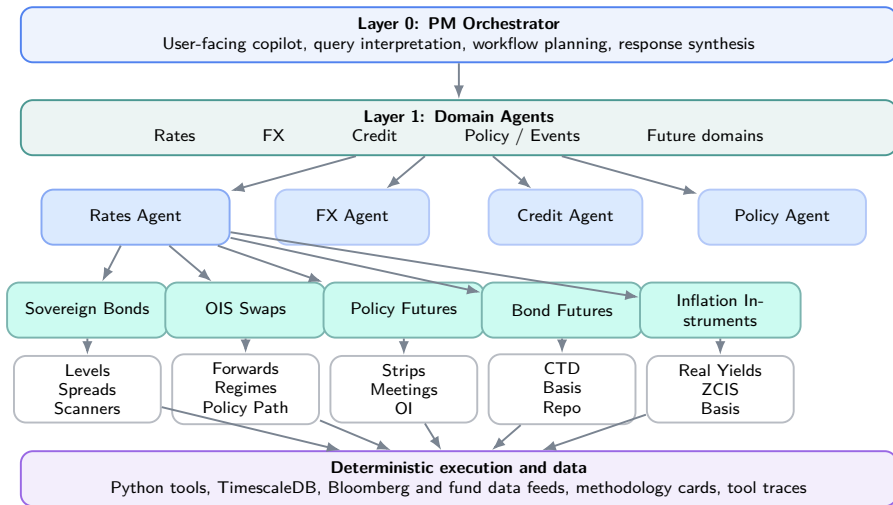
Product

A PM asks in market language. The orchestrator maps intent to deterministic tools. Python computes the numbers against controlled data. The copilot returns a concise synthesis with tool provenance.

Why it is not generic chat

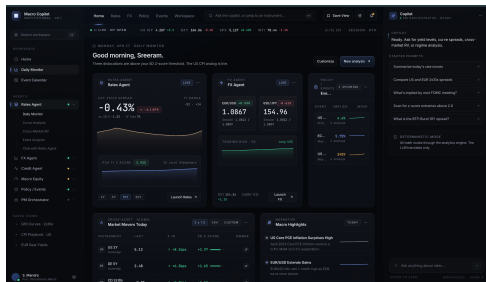
- ▶ The LLM routes and narrates. It does not do market math.
- ▶ Tool traces expose data source, method, and assumptions.
- ▶ Bucket 1A, 1B, and 2 separate deterministic math from parameterized and model-based analytics.
- ▶ The product augments PM judgement. It does not recommend trades.

Architecture: deterministic-first workflow execution



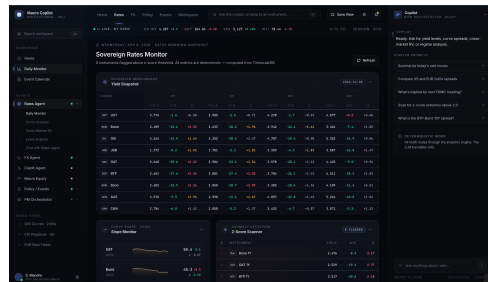
Product surface: dashboard and live market monitor

Daily Monitor



Home view for morning snapshot, dislocation cards, cross-asset context, and quick launch into deeper analysis.

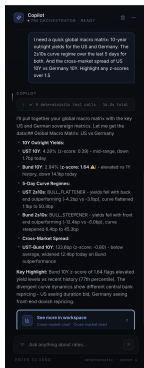
Sovereign Rates Monitor



Rates page with benchmark yields, daily moves, z-scores, curve shape, and anomaly detection.

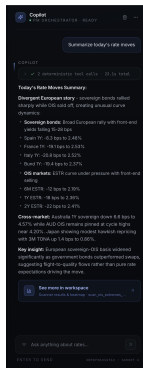
Copilot: natural language workflow execution

Example multi-step query



A PM asks for yields, curve regimes, cross-market spread, and z-score highlights in one conversational turn.

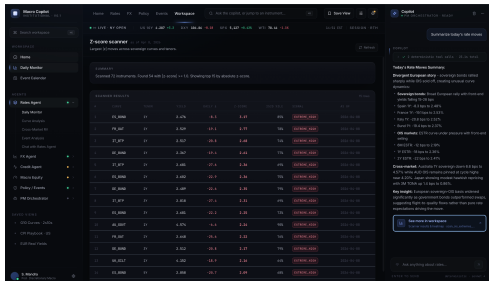
Morning rate-move summary



The response shows deterministic tool-call count, concise synthesis, and workspace drill-down.

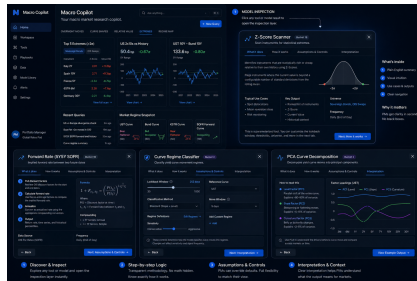
Workspace and model inspection

Workspace scanner



A flexible workspace for scanners, heatmaps, charts, and saved analytical views.

Tool inspection layer



Methodology tabs expose what the tool does, how it works, assumptions, controls, and interpretation.

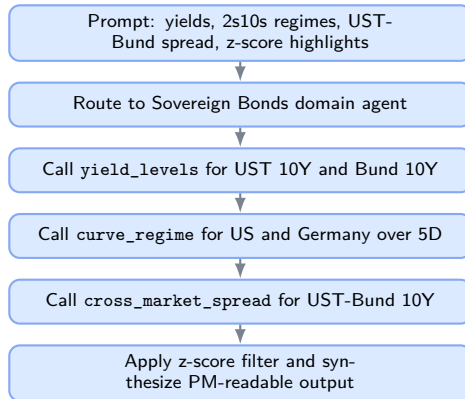
Tool taxonomy and example workflow

Internal tool taxonomy

- 1A** **Strict deterministic core.** One correct answer given data and fixed conventions. Examples: levels, spreads, simple forwards, scanners with house defaults.
- 1B** **Parameterized deterministic sandbox.** Same math, but user-selected assumptions. Examples: custom z-scores, carry and roll, curve fitting, meeting pricing, historical replay.
- 2** **Statistical and model-based engines.** Estimated, probabilistic, or latent-factor outputs. Examples: PCA, HMM, term-premium models, cointegration, GARCH.

Rule: the LLM routes and narrates; deterministic tools compute; assumptions remain visible.

Example workflow: US vs Germany matrix



Rates universe: product catalogue overview

Instrument family	Status	Macro role	Representative analytics
Sovereign Bonds	LIVE	Nominal duration, curve shape, cross-market RV, safe-asset stress	Yield levels, 2s10s, 5s30s, butterflies, UST-Bund, BTP-Bund, scanners, PCA.
OIS Swaps	LIVE	Risk-neutral policy path and clean curve layer	OIS levels, forwards, SOFR-ESTR, OIS regimes, per-meeting policy path.
Policy Futures / STIR	SPEC	Exchange-traded front-end pricing, volume, OI, event repricing	Implied rates, strips, packs, futures-OIS basis, meeting pricing.
Government Bond Futures	SPEC	Liquid duration, CTD mechanics, cash-futures basis, positioning	Futures prices, volume/OI, CTD, implied repo, net basis, basis monitor.
Inflation-Indexed Bonds	PLANNED	Real yields, breakevens, linker carry, inflation-linked duration	Real-yield levels, breakevens, 5Y5Y, linker scanners, carry.
Inflation Swaps	PLANNED	Derivative inflation compensation and forward inflation	ZCIS rates, forwards, cross-market spreads, swap-breakeven basis, CPI fixing path.

Built today: Sovereign Bonds and OIS

are live in the MVP

Specified next: futures and inflation

domains have playbooks and tool specs.

Longer-run edge: 1B and Bucket 2

build the quant analyst layer.

Catalogue: Sovereign Bonds

Macro role: nominal sovereign curves are the cash-market benchmark for duration, collateral, fiscal credibility, and cross-market RV.

Status: **LIVE** six tools live in MVP.

Tool	Bkt	Input to output	Why PMs care
yield_levels	1A	Country, tenor, date to yield, changes, percentile, z-score	Outright rate context for morning checks.
curve_spread	1A	Country and two tenors to 2s10s, 5s30s, history, z-score	Curve steepening and flattening.
cross_market_spread	1A	Two country-tenor pairs to UST-Bund, BTP-Bund, OAT-Bund	Macro divergence and euro periphery stress.
butterfly	1A	Three tenors and weights to curvature and wing split	Belly richness and curve-shape RV.
scanner	1A	Universe and threshold to ranked z-score extremes	Finds stretched instruments quickly.
curve_regime	1B	Curve, lookback, thresholds to bull or bear steepener or flattener	Names the move and gives regime context.
duration_dv01	1A	Bond and date to duration, DV01, convexity	Risk sizing and hedge ratios.
carry_and_roll	1B	Bond, horizon, financing to carry and roll-down	Separates expected carry from directional view.
asset_swap_spread	1B	Bond, swap curve, convention to ASW, Z-spread, basis	Cash-vs-swap RV and balance-sheet stress.
pca_yield_curve / term_premium_acm	2	Curve history and model spec to factors or premium decomposition	Separates level, slope, curvature, expected rates, and duration premium.

Catalogue: Overnight Index Swaps

Macro role: OIS curves are the cleanest traded layer for risk-neutral central-bank path pricing and collateralized derivatives discounting.

Status: **LIVE** five tools live in MVP.

Tool	Bkt	Input to output	Why PMs care
rate_level	1A	Curve, tenor, date to OIS rate, moves, z-score	Clean policy-linked rate snapshot.
curve_spread	1A	One curve and two tenors to OIS slope	Clean steepener or flattener read.
forward_rate	1A	Start and end tenors to 1Y1Y, 2Y1Y, 5Y5Y	Terminal-rate and long-run nominal policy proxies.
cross_market_spread	1A	Two curves and same tenor to SOFR-ESTR, SOFR-SONIA	Cross-central-bank divergence.
scanner	1A	OIS universe and threshold to ranked extremes	Flags stretched policy-pricing nodes.
zero_curve_bootstrap	1B	Pillars, calendars, interpolation to discount factors and zero curve	Exact forwards, DV01, scenarios.
per_meeting_pricing	1B	OIS strip, meeting calendar, stubs to implied hikes/cuts	Core central-bank pricing workflow.
policy_path_since_event	1B	Event date and meeting path to pre/post repricing	Shows what changed after CPI, FOMC, ECB, payrolls.
ois_vs_stir_basis	1B	OIS and futures strips to basis by period	Reconciles OTC and exchange-traded pricing.
pca_ois_curve / term_premium_ois_acm	2	OIS history and model spec to factors and premium decomposition	Model-based policy path and long-end premium split.

Catalogue: Policy Futures / STIR Futures

Macro role: STIR futures are exchange-traded front-end rates instruments. SOFR and SONIA reference RFRs. Euribor references unsecured 3M Euribor. Status: **SPEC** playbook specified.

Tool	Bkt	Input to output	Why PMs care
futures_price_level	1A	Curve, strip position, date to price, moves, z-score, OI	Front-end market monitor.
futures_implied_rate	1A	Futures price to implied rate equals 100 minus price	Prevents sign mistakes in rate-equivalent moves.
futures_strip_snapshot	1A	Curve and date to first 8 contracts with price, rate, volume, OI	Morning strip view.
calendar_spread / butterfly	1A	Two or three contracts to implied-rate slope or curvature	Front-end curve shape.
futures_scanner	1A	Strip universe to price, rate, volume, and OI extremes	Finds event-driven repricing and crowded nodes.
roll_adjusted_history	1B	Actual contracts and roll rule to continuous series	Backtest-grade futures history.
contract_metadata	1B	Generic and actual contract data to expiry, accrual, serial or quarterly	Prevents wrong whites/reds pack logic.
policy_path_from_futures	1B	Futures strip, basis, meetings, known fixings to implied path	Converts exchange prices to central-bank path.
per_meeting_pricing_fut	1B	Meetings, step size, fixings, basis to implied cuts or hikes	Fed, ECB, BoE meeting-pricing workflow.
terminal_rate_distribution	2	Futures, options, model spec to probability distribution	Model-based terminal-rate scenarios.

Catalogue: Government Bond Futures

Macro role: bond futures are the most liquid expression of sovereign duration, with CTD, repo, basis, roll, and positioning layers.

Status: **SPEC** 19 contracts across 9 countries.

Tool	Bkt	Input to output	Why PMs care
futures_price_level	1A	Contract and date to price, changes, z-score	Liquid duration monitor.
futures_volume_oi	1A	Contract and date to volume, OI, changes	Flow and positioning proxy.
scanner_futures	1A	Contract universe to ranked price extremes	Cross-market futures triage.
inter_commodity_spread	1A	Two contracts to raw spread	Dashboard view of curve or country spreads.
ctd_identifier	1B	Basket, conversion factors, prices, repo to ranked CTD list	Foundation for every basis tool.
implied_repo_rate	1B	Future, CTD, cash price to implied financing rate	Basis-trade economics.
gross_basis / net_basis	1B	Future, deliverable bond, repo, carry to basis measures	Cash-futures rich-cheap and stress signals.
futures_dv01	1B	Contract and CTD to DV01 per contract	Hedge ratios and curve spreads.
basis_trade_monitor	1B	CTD, repo, futures, OI to basis, IRR, stress flags	Monitors levered RV fragility.
quality_option_valuation	2	Basket, vol model, rates dynamics to delivery-option value	CTD switching and optionality.

Catalogue: Inflation-Indexed Bonds

Macro role: linkers split nominal yields into real yields and inflation compensation for real-rate and breakeven workflows.

Status: **PLANNED** 24 generic real-yield benchmarks.

Tool	Bkt	Input to output	Why PMs care
real_yield_level	1A	Linker curve, tenor, date to real yield, changes, z-score	Real cost of capital monitor.
breakeven_simple	1A	Nominal generic yield and real generic yield to breakeven	Inflation compensation dashboard.
forward_breakeven	1A	5Y and 10Y breakevens to 5Y5Y forward breakeven	Long-run inflation-compensation anchor.
real_yield_curve_spread	1A	Two real-yield tenors to real curve slope	Real-rate curve trades.
scanner_linkers	1A	Linker and breakeven universe to ranked extremes	Finds real-yield and breakeven dislocations.
tips_carry_seasonal	1B	CPI seasonality, index ratios, horizon to expected carry	Timing linker-vs-nominal trades.
exact_matched_breakeven	1B	Specific nominal and linker bonds to bond-level breakeven	Avoids generic maturity mismatch.
uk_rpi_cpih_monitor	1B	UK linker curve and maturity dates to reform wedge	Handles 2030 RPI methodology break.
swap_minus_be_basis	1B	ZCIS and breakevens to ZCIS minus bond breakeven	Linker cheapness vs swaps.
inflation_risk_premium / hmm_inflation_regime	2	Breakevens, real yields, CPI surprises to premia or regimes	Distinguishes expectations, risk premium, and inflation state.

Catalogue: Inflation Swaps

Macro role: ZC inflation swaps are the derivative-market inflation compensation curve and main comparator to linker breakevens. Status: **PLANNED** 21 USD, EUR, GBP ZCIS benchmarks.

Tool	Bkt	Input to output	Why PMs care
<code>inflation_swap_rate</code>	1A	Curve, tenor, date to ZCIS rate, moves, z-score	Derivative inflation-compensation level.
<code>inflation_curve_spread</code>	1A	Curve and two tenors to spot curve slope	Front-end vs long-end inflation repricing.
<code>inflation_swap_forward</code>	1A	Two spot tenors to forward ZCIS, e.g. 5Y5Y	Long-run inflation compensation.
<code>cross_market_swap_spread</code>	1A	Two curves and tenor to USD-EUR or GBP-EUR spread	Cross-market inflation RV.
<code>inflation_swap_scanner</code>	1A	ZCIS universe to ranked extremes	Flags inflation-swap dislocations.
<code>swap_be_basis_simple</code>	1A	ZCIS and generic breakeven to ZCIS minus breakeven	Basis between swaps and linkers.
<code>curve_builder</code>	1B	Pillars, interpolation, index rules to smooth curve	Off-pillar forwards and pricing.
<code>cpi_fixing_path</code>	1B	Front-end swaps and assumptions to implied monthly CPI path	CPI fixing and event workflow.
<code>uk_rpi_transition</code>	1B	GBP ZCIS, dates, RPI reform to reform exposure	Structural UK inflation caveat.
<code>risk_premium_decomp</code> <code>/ multi_market_pca</code>	2	ZCIS panels, surveys, model spec to premia or factors	Inflation risk premium and global inflation factor map.

Complete tool catalogue appendix

Purpose of this appendix

Every tool listed in Section 9 of the internal instrument files is included below. Descriptions are compressed for external readability; the internal Markdown files remain the methodology source of record.

Instrument family	Tool count	Coverage
Sovereign Bonds	33	nominal yield levels, curve shape, cross-market RV, carry, curve fitting, term premium, flows
OIS Swaps	31	RFR levels, OIS forwards, policy path, zero curves, cross-CB spreads, regimes, term premia
Government Bond Futures	31	futures levels, volume/OI, CTD, implied repo, basis, DV01, positioning, quality option
Inflation-Indexed Bonds	25	real yields, breakevens, forward breakevens, linker carry, inflation risk premia, regimes
Inflation Swaps	30	ZCIS levels, forward inflation, swap-breakeven basis, CPI fixing paths, inflation premia
Policy Futures / STIR Futures	31	SOFR, Euribor, SONIA futures strips, implied rates, packs, event repricing, meeting pricing

Complete tools: Sovereign Bonds (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
yield_levels	Built	1A	takes country, tenor, and date; queries the daily yields table for th...
curve_spread	Built	1A	takes country and two tenors; computes yield differential at each his...
cross_market_spread	Built	1A	takes two (country, tenor) pairs; computes yield differential time se...
butterfly	Built	1A	takes country and three tenors with optional weights (default 50-50)...
curve_regime	Built	1A	takes country and lookback windows (1d/5d/22d); classifies the curve...
scanner	Built	1A	takes z-score threshold (default 2.0) and tenor universe; iterates yi...
yield_change_attribution	Planned	1A	takes country, tenor, and date range; decomposes the cumulative yield...
carry_and_roll_simple	Planned	1A	takes instrument, holding horizon (1M/3M), and a financing rate; comp...
duration_dv01	Planned	1A	takes instrument and date; computes Macaulay duration, modified durat...
otr_ofr_spread	Planned	1A	takes country and tenor; computes spread between the OTR and the most...
zscore_custom	Planned	1B	takes series, lookback window, and standardization method (rolling vs...
curve_fitter_ns	Planned	1B	takes set of bond yields at a given date and a model choice (Nelson-S...
rich_cheap_screen	Planned	1B	takes country and date; runs curve_fitter_ns and ranks bonds by resid...
asset_swap_spread	Planned	1B	takes sovereign bond, a swap curve, and a convention choice (par-ASW...
beta_adjusted_spread	Planned	1B	takes two yield series, a regression window (rolling vs expanding), a...
cointegration_test	Planned	1B	takes two or more yield/spread series, lag specification, and test ch...
half_life	Planned	1B	takes residual or spread series; fits an Ornstein-Uhlenbeck process

Complete tools: Sovereign Bonds (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
rolling_regression	Planned	1B	takes target series and one or more regressors with a rolling window...
curve_regime_parameterized	Planned	1B	takes country, lookback windows, and threshold parameters (front-end...
carry_and_roll_curve_fitted	Planned	1B	takes instrument, horizon, financing assumption (custom repo curve or...
parametric_scenario	Planned	1B	takes position spec (instruments and DV01s) and a shock vector (paral...
pca_yield_curve	Planned	2	takes country (or set of countries), tenor universe, and lookback win...
hmm_curve_regime	Planned	2	takes country, feature vector (e.g., daily yield change, slope change...
term_premium_acm	Planned	2	takes country and yield curve history; estimates the ACM five-factor...
dynamic_nelson_siegel	Planned	2	takes country, yield curve history, and lookback; estimates Diebold-L...
garch_yield_vol	Planned	2	takes yield or spread series, model order (typically GARCH(1,1)), and...
historical_replay	Planned	2	takes position spec and a historical window (e.g., 2013-05-01 to 2013...
multi_country_pca	Planned	2	takes set of countries and tenors; computes PCA across the joint cros...
issuance_impact_model	Planned	2	takes country and auction schedule; runs an event-study or regression...
repo_adjusted_yield	Aspirational	ASP	needs repo data we don't ingest.
dealer_positioning_model	Aspirational	ASP	needs Fed primary dealer survey data and TRACE-like sovereign trade d...
survey_vs_market_decomposition	Aspirational	ASP	needs SPF / Bloomberg consensus expectations data.
flow_attribution	Aspirational	ASP	needs Treasury TIC data, pension fund flow data, ETF flow data.

Complete tools: OIS Swaps (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
rate_level	Built	1A	takes curve_family, tenor, and date; queries the daily rates table
curve_spread	Built	1A	(OIS) takes curve_family and two tenors; computes OIS rate differenti...
forward_rate	Built	1A	takes curve_family, a forward window (start/end tenors, or custom dat...
cross_market_spread	Built	1A	(OIS) takes two (curve_family, tenor) pairs; computes OIS-rate differ...
scanner	Built	1A	(OIS) takes z-score threshold (default 2.0) and tenor universe; itera...
butterfly	Planned	1A	(OIS) takes curve_family and three tenors with optional weights (defa...
curve_regime	Planned	1A	(OIS) takes curve_family and lookback windows; classifies the OIS cur...
zero_curve_bootstrap	Planned	1A	takes pillar grid of OIS rates for a curve_family at a date plus the...
libor_ois_or_sofr_term_spread	Planned	1A	takes term-RFR series (Term SOFR or Term SONIA) and the matched-tenor...
zscore_custom	Planned	1B	takes series, lookback window, and standardization method; computes z...
butterfly_weighted	Planned	1B	(OIS) takes curve_family, three tenors, and a weighting scheme (50-50...
curve_regime_parameterized	Planned	1B	(OIS) same as 1A curve_regime with PM-overridable thresholds.
forward_rate_parameterized	Planned	1B	takes curve_family, custom forward window (any start/end date or teno...
per_meeting_pricing	Planned	1B	takes curve_family, the central bank's meeting calendar, and optional...
curve_fitter_ois	Planned	1B	takes full pillar grid of OIS rates and a model choice (NS, NSS, cubi...
beta_adjusted_ois_spread	Planned	1B	takes paired OIS series at chosen tenors across two curve_families, r...

Complete tools: OIS Swaps (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
cointegration_test	Planned	1B	takes paired or basket of OIS series (typically front-end OIS like 1Y...
half_life	Planned	1B	takes OIS-spread series (cross-CB or cross-tenor); fits OU; returns h...
rolling_regression	Planned	1B	generic primitive; takes OIS target series and any regressors; rollin...
parametric_scenario	Planned	1B	(OIS) takes position spec (instruments and DV01s) plus a shock vector...
historical_replay	Planned	1B	(OIS) takes position spec and historical window (e.g., 2022-01-01 to...
pca_ois_curve	Planned	2	takes curve_family, tenor universe, and lookback window; computes pri...
hmm_ois_regime	Planned	2	takes curve_family or multi-curve set, feature vector (OIS levels/cha...
term_premium_ois_acm	Planned	2	takes curve_family and its OIS history; estimates ACM-style affine mo...
dynamic_nelson_siegel_ois	Planned	2	takes curve_family OIS history; estimates Diebold-Li dynamic NS via K...
garch_ois_vol	Planned	2	takes OIS-rate series at chosen tenor and model order; estimates cond...
probabilistic_stress_ois	Planned	2	takes position spec and a fitted regime model; generates OIS-curve-pa...
multi_currency_ois_pca	Planned	2	takes full multi-CB OIS panel; computes joint cross-section PCA; retu...
xccy_basis_aware_cross_market_spread	Aspirational	ASP	would integrate XCCY basis quotes (cross-domain from 07_cross_currenc...
swaption_implied_terminal_rate	Aspirational	ASP	would use swaption volatility surface data (from 06_swaptions_and_rat...
policy_path_decomposition_advanced	Aspirational	ASP	would decompose forwards into "Taylor-rule-implied path" + "market de...

Complete tools: Government Bond Futures (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
futures_price_level	Built	1A	takes (curve_family, contract_code, date) plus history; queries the d...
futures_volume_oi	Built	1A	takes (curve_family, contract_code, date) plus history; returns daily...
scanner_futures	Built	1A	takes z-score threshold (default 2.0) and contract universe; iterates...
inter_commodity_spread	Planned	1A	(raw 1:1) takes curve_family and two contract_codes; computes price d...
positioning_snapshot	Planned	1A	takes contract or contract universe and a positioning data source (CF...
zscore_custom	Planned	1B	generic primitive (same as in OIS/sovereign).
ctd_identifier	Planned	1B	takes contract (curve_family, contract_code, contract month), the del...
implied_repo_rate	Planned	1B	takes contract and CTD identity; returns implied repo for the CTD, an...
gross_basis	Planned	1B	takes contract and a deliverable bond; returns cash_bond_price - futu...
net_basis	Planned	1B	takes contract, a deliverable bond, the term repo rate, and time-to-d...
futures_dv01	Planned	1B	takes contract and CTD identity; computes DV01 either as DV01_CTD / c...
futures_dv01_on_ois	Planned	1B	takes contract, CTD identity, and the OIS zero curve (02_ois_swaps.md...
ctd_yield	Planned	1B	takes contract; identifies CTD via ctd_identifier
inter_commodity_spread_dv01_weighted	Planned	1B	takes curve_family and two contract_codes; uses futures_dv01 for each...
cross_country_futures_spread_dv01_fx_adjusted	Planned	1B	takes two contracts from different currency complexes plus FX spot (0...
calendar_roll_analytics	Planned	1B	takes front and next-deferred contracts and historical roll patterns...

Complete tools: Government Bond Futures (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
futures_curve_regime	Planned	1B	takes country's bond-future complex (e.g., TU/FV/TY/UXY/US/WN for US)...
positioning_extremes_scanner	Planned	1B	takes positioning data per contract plus historical lookback; flags c...
stress_replay_futures	Planned	1B	takes position spec (contracts + sizes) and a historical window (e.g....
basis_trade_monitor	Planned	1B	takes contract; uses ctd_identifier, implied_repo_rate, term repo rat...
pca_futures_curve	Planned	2	per-country bond-future-price-change panel; lookback 1-10Y configurab...
hmm_futures_regime	Planned	2	takes contract or country complex, feature vector (price changes, bas...
cointegration_test_futures	Planned	2	takes paired bond-future series (typically cross-country, e.g., TY-RX...
half_life_ou_futures	Planned	2	takes futures-spread series; OU fit; returns half-life and current de...
garch_futures_vol	Planned	2	takes futures-price series at chosen contract; GARCH fit; returns con...
quality_option_valuation	Planned	2	takes contract, full deliverable basket, calibrated short-rate model...
multi_country_futures_pca	Planned	2	takes full bond-future panel across countries; computes joint cross-s...
futures_basis_predictive_model	Planned	2	beyond descriptive basis_trade_monitor, fits a model relating basis s...
cross_asset_basis_macro_indicator	Aspirational	ASP	would aggregate basis stress signals across UST, Bund, gilt, JGB into...
futures_implied_term_premium	Aspirational	ASP	would extract term-premium-style decomposition from a country's bond...
positioning_flow_decomposition	Aspirational	ASP	beyond raw COT data, would map positioning changes to specific trade...

Complete tools: Inflation-Indexed Bonds (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
real_yield_level	Planned	1A	takes (curve_family, tenor, date); queries the linker daily yields table
breakeven_inflation	Planned	1A	takes (country, tenor, date); joins the nominal yield (from sovereign...
forward_breakeven	Planned	1A	takes two breakeven points (e.g., 5Y and 10Y) and computes forward br...
breakeven_curve_spread	Planned	1A	takes country and two tenors of breakevens; computes differential
real_yield_curve_spread	Planned	1A	takes country and two tenors of real yields; computes spread; returns...
cross_country_breakeven_spread	Planned	1A	takes two countries at the same tenor; computes breakeven differential
scanner_linkers	Planned	1A	takes z-score threshold and the linker universe; iterates real_yield_...
zscore_custom	Planned	1B	generic primitive (same as in sovereign/OIS/futures modules).
tips_carry_seasonal	Planned	1B	takes (TIPS ticker, holding horizon, CPI seasonal model choice); uses...
carry_adjusted_breakeven	Planned	1B	takes (country, tenor, holding period, CPI seasonal model); adjusts t...
breakeven_curve_fitter	Planned	1B	takes breakeven term structure at a date and a model choice (Nelson-S...
real_yield_curve_fitter	Planned	1B	same as above, applied to real yields.
beta_adjusted_breakeven_spread	Planned	1B	takes two breakeven series (e.g., US 10Y vs UK 10Y); runs rolling OLS

Complete tools: Inflation-Indexed Bonds (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
uk_rpi_cpih_basis_monitor	Planned	1B	takes UK linker breakeven curve; identifies the 2030 maturity boundary
breakeven_vs_inflation_swap_basis	Planned	1B	takes tenor and country; computes difference between bond-implied bre...
pca_real_yield_curve	Planned	2	takes country, tenor set, and lookback; computes PCA on real yield ch...
pca_breakeven_curve	Planned	2	takes country, tenor set, and lookback; computes PCA on breakeven cha...
hmm_inflation_regime	Planned	2	takes feature vector (breakeven changes, real yield changes, CPI surp...
inflation_risk_premium_decomposition	Planned	2	takes breakeven term structure and a model spec (DKW or similar affin...
garch_breakeven_vol	Planned	2	takes breakeven series; GARCH fit; returns conditional vol forecast...
cointegration_breakeven_swap	Planned	2	takes bond-implied breakeven and inflation swap rate series; tests co...
historical_replay_linkers	Planned	2	takes linker/breakeven position spec and a historical window (e.g., 2...
inflation_swap_curve	Aspirational	ASP	needs zero-coupon inflation swap data we don't ingest.
cpi_nowcast_carry	Aspirational	ASP	would combine high-frequency CPI component data (gasoline, food, shel...
deflation_floor_valuation	Aspirational	ASP	would price the embedded deflation floor option in TIPS using a stoch...

Complete tools: Inflation Swaps (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
<code>inflation_swap_rate_level</code>	Planned	1A	takes <code>curve_family</code> , <code>tenor</code> , and <code>date</code> ; queries <code>swap_rate_mid</code> / <code>current...</code>
<code>inflation_swap_curve_spread</code>	Planned	1A	takes one curve family and two tenors; computes <code>tenor spread</code> , <code>period...</code>
<code>inflation_swap_forward</code>	Planned	1A	takes one curve family and a forward window defined by two tenors, e.g.
<code>cross_market_inflation_swap_spread</code>	Planned	1A	takes two curve families and a tenor or forward window; computes <code>raw...</code>
<code>inflation_swap_scanner</code>	Planned	1A	scans all USD/EUR/GBP ZCIS pillars for z-score extremes above thresho...
<code>swap_breakeven_basis_simple</code>	Planned	1A	takes curve family, country/linker family, <code>tenor</code> , and <code>date</code> ; computes...
<code>inflation_swap_event_change</code>	Planned	1A	takes curve family, event date, and tenors; computes <code>pre/post changes...</code>
<code>zscore_custom</code>	Planned	1B	takes any inflation swap series, <code>lookback window</code> , and <code>standardization...</code>
<code>forward_inflation_parameterized</code>	Planned	1B	computes forwards for arbitrary windows with user-selected compoundin...
<code>inflation_curve_builder</code>	Planned	1B	takes ZCIS pillars and curve construction parameters; builds a smooth...
<code>seasonality_adjusted_front_end</code>	Planned	1B	takes curve family, front-end <code>tenor</code> , known inflation prints, <code>release...</code>
<code>cpi_fixing_path</code>	Planned	1B	takes index family, <code>start date</code> , <code>end date</code> , known prints, and <code>assumptio...</code>
<code>swap_breakeven_basis_exact</code>	Planned	1B	takes ZCIS curve, selected nominal/linker bonds or generic curves, <code>ma...</code>
<code>uk_rpi_cpih_transition_monitor</code>	Planned	1B	takes GBP ZCIS curve and RPI reform metadata; estimates the <code>pre/post...</code>
<code>cross_market_index_adjusted_spread</code>	Planned	1B	adjusts raw USD/EUR/GBP inflation swap spreads for index-basket diffe...

Complete tools: Inflation Swaps (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
<code>inflation_swap_carry_roll</code>	Planned	1B	computes expected carry and roll for a ZCIS position over a user-sele...
<code>parametric_scenario_inflation_swaps</code>	Planned	1B	applies user-defined shocks to ZCIS curves or forwards; returns rate...
<code>historical_replay_inflation_swaps</code>	Planned	1B	replays historical inflation swap curve paths through selected positi...
<code>pca_inflation_swap_curve</code>	Planned	2	takes curve family, tenor set, and lookback; computes PCA on ZCIS cha...
<code>multi_market_inflation_pca</code>	Planned	2	computes joint PCA across USD/EUR/GBP curves; separates global inflat...
<code>inflation_risk_premium_decomposition_swaps</code>	Planned	2	estimates expected inflation vs inflation risk premium from ZCIS usin...
<code>hmm_inflation_swap_regime</code>	Planned	2	fits HMM or other regime model to ZCIS rates/forwards, breakevens, re...
<code>garch_inflation_swap_vol</code>	Planned	2	estimates conditional volatility of inflation swap changes at selecte...
<code>cointegration_swap_breakeven_basis</code>	Planned	2	tests whether ZCIS and bond breakevens share a stable long-run relati...
<code>probabilistic_inflation_path</code>	Planned	2	generates a distribution of CPI/RPI/HICP outcomes consistent with inf...
<code>yoy_inflation_swap_curve</code>	Aspirational	ASP	requires YoY inflation swap data.
<code>inflation_options_surface</code>	Aspirational	ASP	caps/floors and inflation swaptions; used to infer inflation distribu...
<code>cpi_fixing_swaps</code>	Aspirational	ASP	requires short-dated fixing instruments; valuable for CPI release and...
<code>sdr_liquidity_monitor</code>	Aspirational	ASP	uses SDR/EMIR/DTCC trade repository data where available to estimate...
<code>real_time_cpi_nowcast_overlay</code>	Aspirational	ASP	combines high-frequency energy, food, rent, used-car, and other compo...

Complete tools: Policy Futures / STIR Futures (1/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
futures_price_level	Planned	1A	takes curve_family, strip_position, and date; returns last price, 1d/...
futures_implied_rate	Planned	1A	takes futures price and inverse_pricing flag; computes implied_rate =...
futures_strip_snapshot	Planned	1A	takes curve family and date; returns first 8 contracts with price, im...
futures_calendar_spread	Planned	1A	takes curve family and two strip positions; computes implied-rate spr...
futures_butterfly_simple	Planned	1A	takes three strip positions with default 50-50 weights; computes stri...
futures_pack_average_simple	Planned	1A	takes fixed contract group, e.g.
futures_cross_market_spread	Planned	1A	takes two curve families and matched strip positions; computes cross...
volume_open_interest_snapshot	Planned	1A	returns daily volume, open interest, and 1d changes by contract.
futures_scanner	Planned	1A	scans price moves, implied-rate moves, volume z-scores, and OI z-scor...
zscore_custom	Planned	1B	generic primitive allowing lookback, standardization method, and roll...
roll_adjusted_futures_history	Planned	1B	takes actual-contract history and roll rule; constructs a continuous...
futures_strip_with_contract_metadata	Planned	1B	joins rolling generics to actual contract-month metadata, accrual per...
policy_path_from_futures	Planned	1B	takes futures strip, meeting calendar, policy target series, benchmar...
per_meeting_pricing_from_futures	Planned	1B	decomposes futures-implied rates into expected per-meeting hike/cut a...
event_repricing_futures	Planned	1B	takes event date/time and curve family; computes pre/post changes by...
futures_ois_basis	Planned	1B	compares futures-implied rates with OIS-derived forward rates over ma...

Complete tools: Policy Futures / STIR Futures (2/2)

Every row is sourced from Section 9 of the internal instrument file. Descriptions are shortened for catalogue readability.

Tool	Status	Bkt	Catalogue description
convexity_adjusted_futures_forward	Planned	1B	applies a selected convexity/margining adjustment to futures-implied...
pack_bundle_analytics	Planned	1B	creates custom packs/bundles with simple, DV01/tick-value, or user-sp...
positioning_zscore	Planned	1B	computes volume/open-interest z-scores with roll-window handling and...
historical_replay_policy_futures	Planned	1B	takes futures position vector and historical window; replays strip mo...
pca_policy_futures_strip	Planned	2	computes PCA on implied-rate changes across the futures strip.
hmm_policy_pricing_regime	Planned	2	fits a regime model on strip levels/slopes, OIS forwards, realized vo...
central_bank_reaction_function_model	Planned	2	estimates how futures-implied policy paths respond to macro surprises...
terminal_rate_distribution	Planned	2	combines futures, OIS, and optionally options to estimate a distribut...
positioning_crowding_model	Planned	2	uses volume, open interest, price action, and roll behavior to infer...
garch_front_end_futures_vol	Planned	2	estimates conditional volatility for front-end futures-implied-rate c...
multi_market_policy_pca	Planned	2	PCA across SOFR, Euribor, and SONIA futures panels to identify global...
fedwatch_style_probabilities	Aspirational	ASP	true FedWatch-style probabilities require 30-Day Fed Funds futures, n...
intraday_event_repricing	Aspirational	ASP	requires intraday futures data around CPI, payrolls, and central-bank...
futures_options_terminal_distribution	Aspirational	ASP	requires SOFR/SONIA/Euribor futures options or mid-curve options data.
dealer_positioning_inference	Aspirational	ASP	would require richer positioning data, CFTC-style reports, client-flo...